

Logical Reasoning – Letter Series

Tricks, patterns and a step-by-step solving strategy for placement and aptitude exams

Letter Series is a common topic in aptitude and reasoning tests, especially in placement exams. The main skill is to identify the pattern between letters and predict the next letter(s).

Examples

A, C, E, G, ? → I

B, E, H, K, ? → N

Z, W, T, Q, ? → N

1. First learn the Alphabet Number System

Memorize these values:

Letter	Value	Letter	Value
A	1	N	14
B	2	O	15
C	3	P	16
D	4	Q	17
E	5	R	18
F	6	S	19
G	7	T	20
H	8	U	21
I	9	V	22
J	10	W	23
K	11	X	24
L	12	Y	25
M	13	Z	26

Important shortcut

Remember: **A=1, E=5, I=9, M=13, Q=17, U=21, Y=25**

These are every 4th letter and help you calculate quickly.

2. Trick 1 - Check the Constant Gap

The easiest pattern is a fixed increase.

Example

A, C, E, G, ?

A = 1, C = 3, E = 5, G = 7

Pattern: +2, +2, +2

$7 + 2 = 9 = \text{I}$

Answer: I

Practice

D, G, J, M, ?

D = 4, G = 7, J = 10, M = 13

Pattern: +3

$13 + 3 = 16 \rightarrow \text{P}$

Answer: P

3. Trick 2 - Decreasing Series

Sometimes the letters move backward.

Example

Z, W, T, Q, ?

Numbers: 26, 23, 20, 17

Pattern: -3, -3, -3

$17 - 3 = 14 \rightarrow \text{N}$

Answer: N

4. Trick 3 - Increasing Gap

Sometimes the gap itself increases.

Example

A, B, D, G, K, ?

Positions: 1, 2, 4, 7, 11

Differences: +1, +2, +3, +4

Next: +5

$11 + 5 = 16 \rightarrow \text{P}$

Answer: P

Important pattern - look for:

+1, +2, +3, +4 ...

+2, +4, +6, +8 ...

+3, +5, +7, +9 ...

5. Trick 4 - Alternating Pattern

This is very important for intermediate and advanced questions.

Example

A, D, C, F, E, H, ?

Separate odd and even positions:

Odd positions: A, C, E, ? $\rightarrow +2$

Even positions: D, F, H $\rightarrow +2$

Odd next = G

Answer: G

Shortcut

If the difference doesn't make sense from left to right, try the 1st, 3rd, 5th, 7th... letters and the 2nd, 4th, 6th, 8th... letters separately.

6. Trick 5 - Two Different Patterns

Example

A, C, F, H, K, M, ?

$A \rightarrow C = +2$, $C \rightarrow F = +3$, $F \rightarrow H = +2$, $H \rightarrow K = +3$, $K \rightarrow M = +2$

Pattern: +2, +3, +2, +3, +2 ...

Next: $M + 3 = P$

Answer: P

7. Trick 6 - Reverse Alphabet

Remember the mirror pairs:

A $\leftrightarrow$ Z	B $\leftrightarrow$ Y	C $\leftrightarrow$ X	D $\leftrightarrow$ W
E $\leftrightarrow$ V	F $\leftrightarrow$ U	G $\leftrightarrow$ T	H $\leftrightarrow$ S
I $\leftrightarrow$ R	J $\leftrightarrow$ Q	K $\leftrightarrow$ P	L $\leftrightarrow$ O
M $\leftrightarrow$ N			

Example

A, Z, B, Y, C, X, ?

Pattern: A, B, C, ? and Z, Y, X

Therefore next = D

Answer: D

8. Trick 7 - Jump Forward and Backward

Example

B, E, D, G, F, I, ?

$B \rightarrow E = +3$, $E \rightarrow D = -1$, $D \rightarrow G = +3$, $G \rightarrow F = -1$, $F \rightarrow I = +3$

Pattern: $+3, -1, +3, -1, +3 \dots$

$I - 1 = H$

Answer: H

9. Trick 8 - Multiplication / Position Pattern

For difficult questions, convert letters into numbers.

Example

B, D, H, P, ?

Positions: 2, 4, 8, 16

Pattern: $\times 2$

Next: $16 \times 2 = 32$

Alphabet has only 26 letters, so wrap around: $32 - 26 = 6 \rightarrow F$

Answer: F

Remember

If the calculation goes beyond 26, subtract 26.

Example: $29 \rightarrow 29 - 26 = 3 \rightarrow C$

10. Trick 9 - Circular Alphabet

The alphabet can continue after Z. Think of it as: ... X, Y, Z, A, B, C, D ...

Example

W, Z, C, F, ?

$W = 23$, $Z = 26$, $C = 3$, $F = 6$

Pattern: $+3$

$W \rightarrow Z \rightarrow C \rightarrow F \rightarrow I$

Answer: I

This technique is extremely useful in advanced questions.

11. Trick 10 - Difference of Differences

For difficult series, don't only look at the first difference.

Example

A, C, F, J, O, ?

Positions: 1, 3, 6, 10, 15

Differences: +2, +3, +4, +5

Next difference: +6

$15 + 6 = 21 \rightarrow \text{U}$

Answer: U

12. Trick 11 - Pair Pattern

Sometimes letters work in pairs.

Example

AB, DE, GH, JK, ?

Each pair moves +3.

Next pair: MN

Answer: MN

13. Trick 12 - Reverse Pair Pattern

Example

ZA, YB, XC, WD, ?

First letters: Z, Y, X, W $\rightarrow$ decreasing by 1

Second letters: A, B, C, D $\rightarrow$ increasing by 1

Therefore: VE

Answer: VE

This type is common in higher-level reasoning.

14. The Most Important Solving Order

When you see a Letter Series question, don't randomly guess the pattern. Use this order:

Step	What to do
1	Convert letters into numbers. Example: C, F, I, L, ? $\rightarrow$ 3, 6, 9, 12, ?
2	Check the differences: +3, +3, +3
3	If that doesn't work, check +1, +2, +3 ...
4	Check alternating positions: 1st, 3rd, 5th ... and 2nd, 4th, 6th ...
5	Check forward/backward combinations such as +3, -1, +3, -1 ...
6	Check reverse alphabet: A $\leftrightarrow$ Z
7	Check circular alphabet: Z $\rightarrow$ A
8	For advanced questions, check: increasing differences, decreasing differences, alternating differences, pair patterns, multiple patterns, circular patterns, position multiplication/division

15. Fast Exam Trick

When you have only 20–30 seconds, use the **5-second method**: look at the first 3–4 letters and ask:

Question	Example
① Is it +same number?	A C E G $\rightarrow$ +2
② Is it -same number?	Z W T Q $\rightarrow$ -3
③ Is the gap increasing?	A B D G K $\rightarrow$ +1, +2, +3, +4
④ Is it alternating?	A D C F E H $\rightarrow$ +3, -1 or +2 pattern on alternate positions
⑤ Is it going through Z $\rightarrow$ A?	W Z C F $\rightarrow$ +3

If none works, split odd/even positions.

16. Common Letter Series Patterns You Should Master

For placement exams, practice these patterns:

Pattern	Example
Constant +	A C E G
Constant –	Z W T Q
Increasing gap	A B D G K
Decreasing gap	Z Y W T P
Alternate	A C B D C E
+2, +3	A C F H K
+3, –1	B E D G F I
Reverse alphabet	Z Y X W
Circular	W Z C F
Odd/Even	A D C F E H
Pair series	AB DE GH JK
Reverse pairs	ZA YB XC WD
Multiplication	B D H P
Difference of differences	A C F J O
Mixed pattern	A Z B Y C X

17. One Golden Rule

Never try to solve a Letter Series only by looking at the letters visually.

Convert A → 1, B → 2, C → 3 ... Z → 26. Then the question becomes a number-series problem.

Example: C, H, M, R, ? becomes 3, 8, 13, 18, ?

Pattern: +5, +5, +5 → 18 + 5 = 23 → W

So the answer is W.